

Milestone II

Adopted Design

Title

Design of Lightweight Broad-Band Microwave Absorbers

Sponsor

Industry Sponsor: Quang Nguyen, DEVCOM

Email : Quang.nguyen7.civ@army.mil

Faculty Sponsor: Dr. Paul Leu

Email: pleu@pitt.edu

Team Members and Roles

Planner	Abegale Jones	aaj43@pitt.edu
Resource Manager	Joseph Antonucci	jpa45@pitt.edu
Presenter & Communicator	Tegan Carr	trc91@pitt.edu

1. Preliminary Design Concepts (Tegan)

Candidate Design Concepts

Multi-Objective Genetic Algorithm (GA) Approach:

Overview: Use a genetic algorithm to simulate evolution, where each "generation" consists of possible material combinations. Each material combination (chromosome) is evaluated for radar absorption effectiveness, and top performers are used to create new combinations. Genetic algorithms are well-suited for exploring a large design space without requiring gradient information, making them useful for complex or noisy search spaces like material combinations.

Bayesian Optimization with Gaussian Processes:

Overview: Bayesian optimization is used to select material combinations based on a probabilistic model that estimates radar absorption performance. This model is updated with each test, gradually "learning" which combinations are likely to perform well. Bayesian optimization is efficient for problems with expensive evaluations because it focuses on promising areas in the search space based on a posterior distribution (often modeled with Gaussian processes).

Gradient Descent with Neural Network Approximation:

Overview: This approach uses a neural network to approximate the radar absorption as a continuous function of the material properties. Gradient descent is then applied to adjust the inputs (material properties) to maximize absorption. This method requires a differentiable model, which a neural network can provide. The network is trained on known material combinations and their absorption values, and gradient descent is used to iteratively improve the selection.

Requirements & Specifications

Multi-Objective Genetic Algorithm (GA) Approach:

Requirement/Specification	Satisfaction by Design
Exploration of large search space	The GA can search extensively across a large number of material combinations.
Non-linear search capability	GAs are adept at handling complex, non-linear relationships between materials and radar absorption.
Avoidance of local optima	GAs naturally avoid local optima through crossover and mutation mechanisms.
Modularity in material combination	GAs can evaluate any combination of three materials from the database, enabling flexibility.
Customizability in optimization parameters	Population size, mutation rate, and crossover can be adjusted based on specific design needs.

Bayesian Optimization with Gaussian Processes:

Requirement/Specification	Satisfaction by Design
Efficient optimization for expensive evaluations	Bayesian optimization reduces evaluations needed to find high-performing combinations, crucial for costly or time-intensive evaluations.
Robustness in noisy data environments	Bayesian models are inherently probabilistic, making them resilient to data noise and measurement variability.
Utilization of prior data	Bayesian optimization can leverage existing material data to accelerate the search for optimal combinations.
Automatic balancing between exploration and exploitation	The probabilistic model enables a balance between exploring new combinations and exploiting known high-performing ones.
Adaptability with smaller data sets	This approach works well with limited data, which is common in new material design spaces.

Gradient Descent with Neural Network Approximation:

Requirement/Specification	Satisfaction by Design
Ability to approximate complex material properties	Neural networks can approximate the function between material properties and radar absorption.
Fast evaluations once trained	The trained neural network provides rapid predictions, supporting fast optimization through gradient descent.
Fine-tuned optimization capability	Gradient descent allows for precise adjustments of material properties to maximize absorption.
Utilization of differentiable model	Neural networks are differentiable, meeting the need for gradient-based optimization.
Support for high-dimensional search spaces	Neural networks can handle multiple input features (material properties), making them suitable for a diverse material database.

Strengths and Weakness

Genetic Algorithm:

Advantages: GAs can explore various combinations broadly, avoiding local optima and finding high-performing combinations in non-linear problem spaces.

Disadvantages: Computationally intensive, especially if the material database is large. Also, tuning parameters like mutation rate and population size can be challenging.

Bayesian Optimization:

Advantages: Works well with smaller datasets and can reduce the number of evaluations needed to find optimal material combinations. Bayesian methods are also robust for noisy data.

Disadvantages: The computational load of updating the probabilistic model can increase as more data is added. Additionally, Gaussian process models may struggle with high-dimensional inputs if the database of materials is large.

Gradient Descent:

Advantages: Effective in leveraging data to guide optimization in complex, high-dimensional spaces. Once trained, the neural network model can provide fast evaluations and allow gradient-based improvements.

Disadvantages: Requires a well-structured, sizable dataset for accurate training. Neural networks can overfit with limited data, and gradient descent might converge to local minima, missing globally optimal material combinations.

Current or Future Supporting Analysis or Validating Experiments

To compare the three algorithms effectively, a cost function can be designed to evaluate their performance based on criteria such as computational efficiency, accuracy in identifying optimal material combinations while remaining below the maximum thickness constraint, and resource requirements for validation.

This cost function combines weighted factors for computational time (favoring algorithms that reach solutions faster), accuracy of radar absorption predictions (measured by deviation from actual physical test results), and experimental cost (number and complexity of physical tests required). For instance, the cost function might assign higher penalties for excessive computational time in the genetic algorithm or high physical testing demands in the gradient descent approach. By applying this cost function, each algorithm can be evaluated based on its overall efficiency and effectiveness in achieving high radar absorption, enabling the team to objectively assess and select the best approach for further development.

Down Select Approach

Most of the down selecting methodology used to determine which algorithm to use comes down to what is illustrated below in the Pugh Chart. Ideally, we would be able to implement all three algorithms and experiment with them by increasing the sample size or the monitoring resource requirements to execute them. However, practically there are time and complexity restraints that make this not feasible. We created five categories to weigh each algorithm against the others. Ultimately, the genetic algorithm and Bayesian optimization proved favorable. We currently have a completed generic algorithm working and implemented; the Bayesian optimization algorithm is in the process of being completed and will be done within the coming weeks.

2. Preliminary Design Review Comments

Reviewing Peer Team

Team 8 reviewed our preliminary design for our python code.

Peer Team Preliminary Design Review Comments

- Add comments to help people understand what each piece of code is doing
- Record the results of each algorithm to compare the materials each algorithm outputs
- Compare the results of the MATLAB code to the python code to ensure consistency

Actions

With the comments we received from Team 8 we decided to ensure that there were ample comments in the code to communicate which each piece was doing. We added comments for each section including heading for functions and specific sections that are used for calculations. Results from each algorithm are also stored using a python script in a folder and sent to GITHUB for further analysis. MATLAB has been used to confirm results as the MATLAB script has been proven true and gives correct results. This will be used to confirm outputs of the various algorithms and to confirm the effectiveness of the algorithm.

3. Pugh Chart

Table 1: Pugh Chart of Preliminary Designs

Criteria	Baseline Script	Optimized GA Code	Gradient Descent Optimization	Bayesian Optimization
Code Efficiency	0	-	+	++
Code Complexity	0	++	-	-
Optimization Capability	0	++	+	++
Development Time	0	-	-	-
Error Handling	0	++	+	+
	Totals	+4	+1	+4

4. Adopted Design

Adopted Design

For our final design we will be using an optimized genetic algorithm code. Code complexity was an important category for use considering the lack of experience each of us have with these types of algorithms. Genetic algorithms have a lot of documentation and open-source functions that can be used effectively, solving the problem of code complexity. Another reason we are choosing genetic algorithm as our primary design is because of the error handling. The time constraint of this project doesn't allow for much debugging so the less time it takes us to debug the more we can work on finding an optimized structure.

Detailed Design Description

We will be implementing a genetic algorithm into MATLAB and Python to compare results. The genetic algorithm will have an objective of finding the lowest reflection of 3 materials being surrounded by air and PEC. It will also have a constraint that the thickness must be less than 7mm. This is to ensure that the material being made is feasible to construct in real life for military applications.

Anticipated Supporting Analysis

To ensure our design meets the requirements and specifications listed above, we will produce a graph of the reflection vs. the frequency of the 3 layered material. We will also display the thickness of each layer and what each layer consists of. Results will be compared to a proven correct MATLAB script to ensure that the material is consistent with proven results.

Sponsor Approval

Our sponsor would like us to try to implement all three algorithms and compare the results of each of them to conclude which of the three algorithms outputs the best results for our 3 layered RAM structure. Because of the time constraint and complexity of this project, an optimized genetic algorithm would be an acceptable deliverable at the end.

5. Updated Project Plan

Refer to teamwork gantt chart to view updated project plan

6. Updated Project Timeline

Refer to teamwork gantt chart to view updated project timeline